9. Message Queue & CDN

Advantage of a message queue..

- Decoupling allows you to seperate components and make them free sooner.
- The system becomes more scalable as more clients can send in request asynchronously.
- You can process the message again, if the first attempt fails. This makes the system fault-tolerant.

Challenges.

- Making sure that the ordering of the messages remains the same.
- Handling duplicate message to avoid redundancy and processing cycle.
- Adding something in the middle introduces latency, the benefits should outweight the lag.

CDN

Content delivery Network

Pros & Cons CDN.

1. Faster load time
2. Reduces latency
3. caching static content.

Challenge

1. Your data should always be fresh

10. CAP Theorem

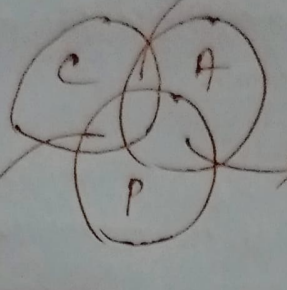
Consistency Availability Partition Tolerence

~~Parat~~ Parameters to evaluate a distributed system.

C – The system should deliver the same results no matter how and from where the query.

A - The system should always return some response or result.

P - ~~Partition~~ Partition tolerance - The system should be functional even if one partition is disconnected.



Ticket booking (C+A)
(P can be sacrificed)

Social Network System (PT+A)
(consistency can be sacrificed)

Banking system
↳ consistency + Partition tolerance
(A may be sacrificed)

Rate Limiter

- Requirement gathering
  - ↳ client side ✗ (Not possible)
  - ↳ server side
  - ↳ API gateway

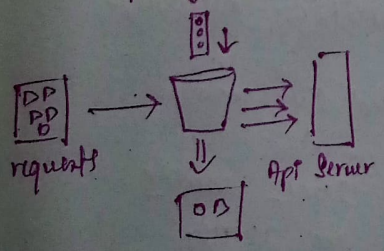
Algorithms

① Token Bucket

② Fixed window counter

③ Sliding window log.

- bucket
- refiller



- easy to implement
- memory efficient
- allows bursts to traffic